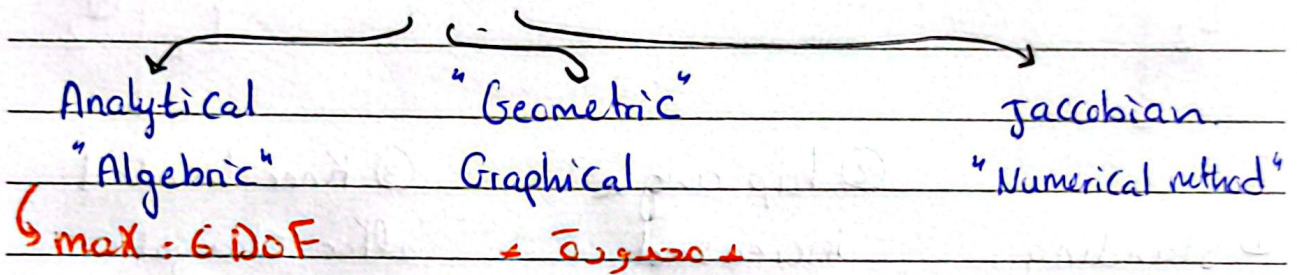


Kinematics

used in real life

	Forward	Inverse
input	$\theta_1, \theta_2, \theta_3$	x, y, z (in 3D)
output	x, y, z	$\theta_1, \theta_2, \theta_3$

Inverse Kinematics methods



Graphical "Geometric" method :

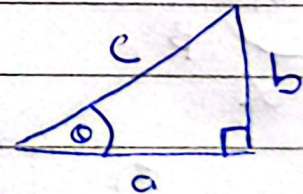
- Solve θ by using triangles
- Trigonometric Formulas for graphical solutions :

① Right angle triangle :

$$a^2 + b^2 = c^2 \rightarrow c = \sqrt{a^2 + b^2}$$

$$\sin \theta = \frac{b}{c}, \cos \theta = \frac{a}{c}$$

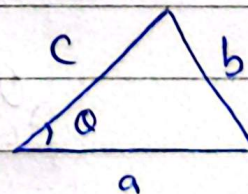
$$\tan \theta = \frac{b}{a}$$



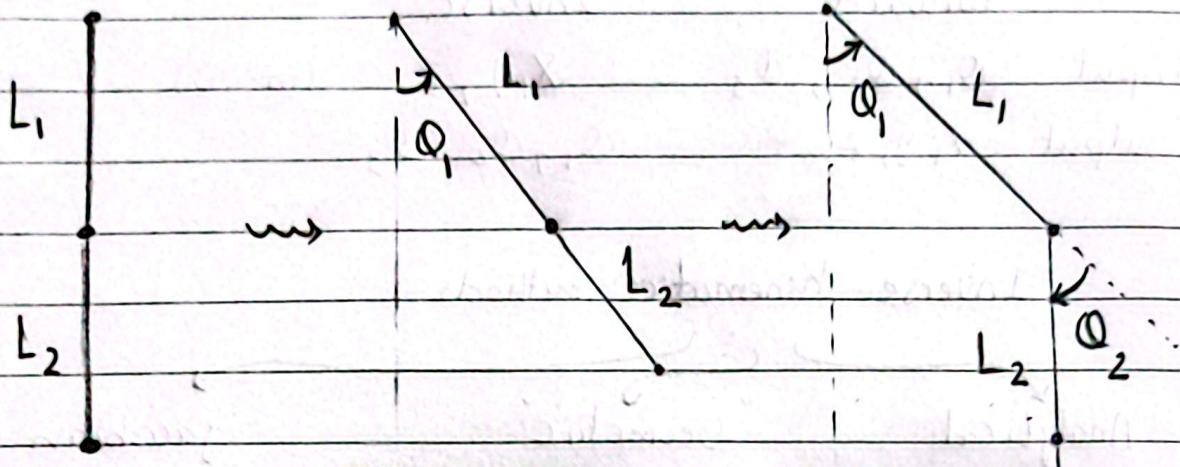
② Law of Cosines :

$$b^2 = a^2 + c^2 - 2ac \cos(\theta)$$

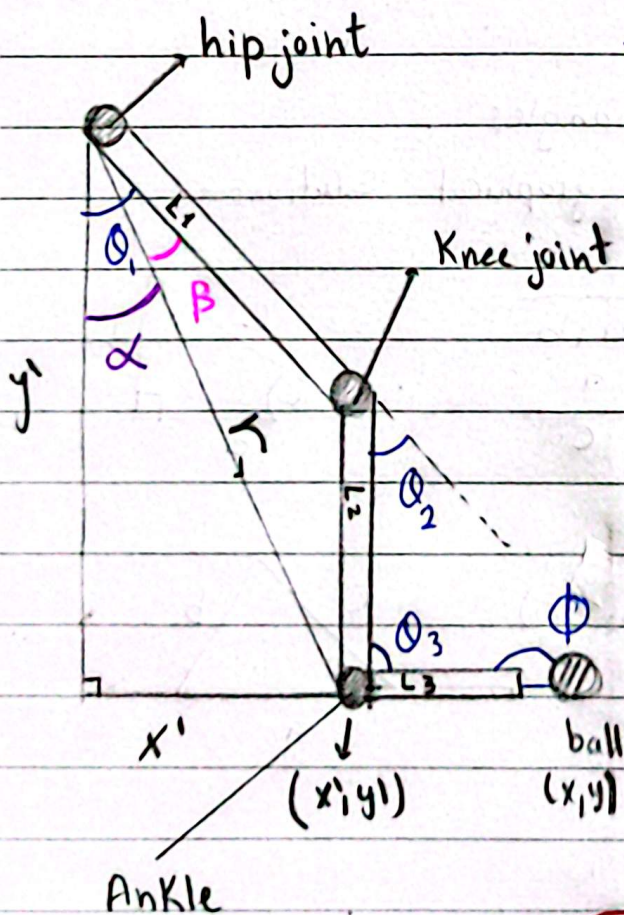
$$\therefore \cos(\theta) = \frac{a^2 + c^2 - b^2}{2ac}$$



Inverse Kinematics in Humanoid Robot.



- ① standing position
- ② hip angle increased
- ③ Knee bended after straight position of leg

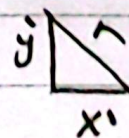


∴ if you want your fingertip to touch the cup, your wrist has to stop a few inches short of the cup
 ∴ (x', y') → the ankle should stop at this position (L_3 distance away), so the foot which is L_3 long perfectly fills that gap.

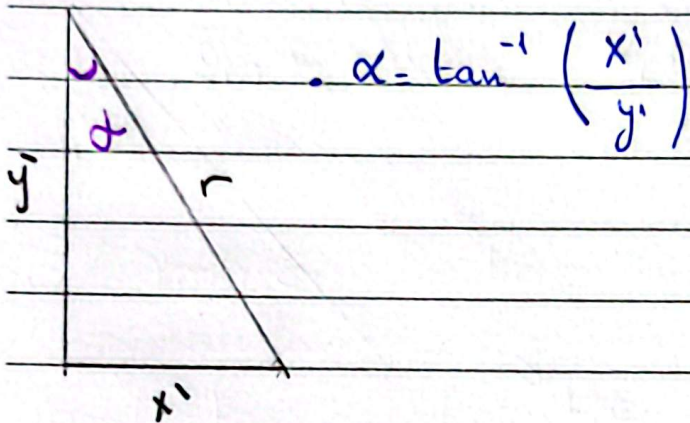
$$x' = x_{ball} - L_3 \cos(\phi)$$

$$y' = y_{ball} - L_3 \sin(\phi)$$

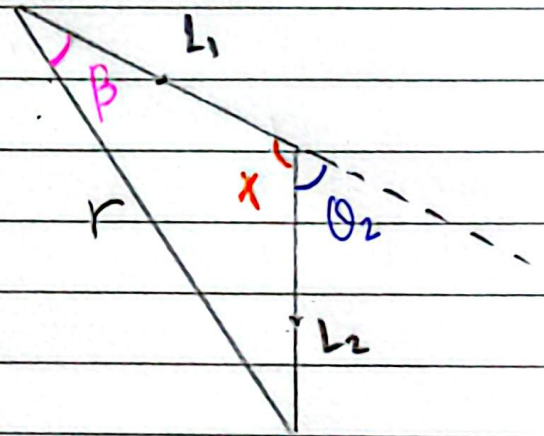
$$r = \sqrt{\dot{x}'^2 + \dot{y}'^2}$$



Triangles and Calculations =



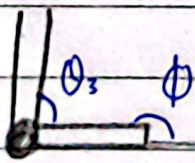
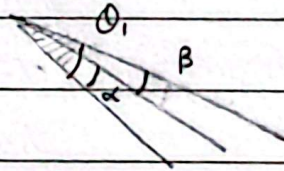
$$\alpha = \tan^{-1} \left(\frac{x'}{y'} \right)$$



$$\beta = \cos^{-1} \left(\frac{L_1^2 + r^2 - L_2^2}{2L_1 r} \right)$$

$$\theta_2 = 180^\circ - x^\circ = 180^\circ - \cos^{-1} \left(\frac{L_1^2 + L_2^2 - r^2}{2L_1 L_2} \right)$$

$$\theta_1 = \alpha + \beta$$



*To keep the foot flat ($\phi = 0^\circ$) while the leg moves, the ankle must compensate for the rotation of the hip and knee.

$$\theta_3 = \phi - (\theta_1 - \theta_2)$$

$$\text{when } \phi = 0^\circ \quad \therefore \theta_3 = \theta_2 - \theta_1$$